CLAIMS

What is claimed is:

1. A surface cleaner comprising:

a housing;

a suction nozzle coupled to the housing;

a recovery tank in fluid communication with the suction nozzle, the recovery tank disposed on the housing in a recovery tank operating position;

a suction source disposed in the housing, the suction source in fluid communication with the suction nozzle and the recovery tank, the suction source configured to draw fluid and debris through the suction nozzle and into the recovery tank when the recovery tank is in the recovery tank operating position;

a supply tank configured to store a cleaning fluid, the supply tank disposed on the housing in a supply tank operating position;

a distributor in fluid communication with the supply tank when the supply tank is in the supply tank operating position, the distributor configured to dispense the cleaning fluid from the supply tank onto a surface being cleaned;

a pump in fluid communication with the supply tank and the distributor, the pump operable to pump the cleaning fluid in a direction from the supply tank to the distributor; and

a pump housing disposed within the housing, the pump disposed in the pump housing and the pump housing including a cooling air inlet and a cooling air outlet in fluid communication with the suction source, and

wherein the housing further includes a first tank guide that facilitates locating at least one of the recovery tank in the recovery tank operating position and the supply tank in the supply tank operating position, the first tank guide including a cooling air inlet in fluid communication with the cooling air inlet of the pump housing.

2. The surface cleaner of claim 1, wherein in the housing further includes a second tank guide that facilitates locating one of the recovery tank in the recovery tank operating position and the supply tank in the supply tank operating position.

3. The surface cleaner of claim 2, wherein the second tank guide forms a cooling air flow path between the cooling air inlet of the first tank guide and the cooling air inlet of the pump housing.

4. The surface cleaner according to claim 3, wherein the first tank guide facilitates locating the supply tank in the supply tank operating position and the second tank guide facilitates locating the supply tank in the supply tank operating position.

5. The surface cleaner according to claim 4, wherein the first tank guide includes a projection that extends outward from the housing, wherein the projection is received in a first recess of the supply tank when the supply tank is in the supply tank operating position, wherein the second tank guide includes a projection that extends outwardly from the housing, wherein the projection of the second tank guide is received in a second recess of the supply tank when the supply tank is in the supply tank operating position.

6. The surface cleaner according to any one of claims 3 through 5, wherein the cooling air inlet of the pump housing is located within the second tank guide.

7. The surface cleaner of claim 6, wherein the pump housing includes an inlet projection that includes the cooling air inlet of the pump housing, wherein the inlet projection extends into the second tank guide.

8. The surface cleaner according to any one of claims 3 through 7, wherein the first tank guide is between the supply tank and the housing when the supply tank is in the supply tank operating position.

9. The surface cleaner according to any one of claims 3 through 8, wherein the cooling air flow path extends through the housing.

10. The surface cleaner according to any one of claims 3 through 9, wherein the housing includes a first side and a second side opposite the first side, wherein the first tank guide is disposed on the first side of the housing and the second tank guide is disposed on the second side of the housing.

11. The surface cleaner according to any one of the preceding claims, wherein the first tank guide includes a projection that extends outwardly from the housing, wherein the projection is received in a recess of one of the recovery tank in the recovery tank operating position and the supply tank in the supply tank operating position.

12. The surface cleaner according to any one of the preceding claims, wherein the housing includes a supply tank tray and a recovery tank tray, wherein the supply tank tray supports the supply tank and the recovery tank tray supports the recovery tank in the supply tank operating position and the recovery tank operating position.

13. The surface cleaner of claim 12, wherein the cooling air inlet is adjacent at least one of the supply tank tray and the recovery tank tray.

14. The surface cleaner according to any one of the preceding claims, further comprising a cooling air duct that extends from the pump housing to a location in a suction airflow path between the recovery tank and the suction source.

15. The surface cleaner according to any one of the preceding claims, further comprising a hose, wherein the suction nozzle is coupled to the housing by the hose and the hose is in fluid communication with the recovery tank.

16. The surface cleaner according to any one of the preceding claims, wherein the first tank guide facilitates locating the supply tank in the supply tank operating position.

17. A surface cleaner comprising:

a housing;

a suction nozzle coupled to the housing;

a recovery tank in fluid communication with the suction nozzle, the recovery tank disposed on the housing in a recovery tank operating position;

a suction source disposed in the housing, the suction source in fluid communication with the suction nozzle and the recovery tank, the suction source configured to draw fluid and debris through the suction nozzle and into the recovery tank when the recovery tank is in the recovery tank operating position;

a supply tank configured to store a cleaning fluid, the supply tank including a first recess and a second recess, the supply tank disposed on the housing in a supply tank operating position;

a distributor in fluid communication with the supply tank when the supply tank is in the supply tank operating position, the distributor configured to dispense the cleaning fluid from the supply tank onto a surface being cleaned;

a pump in fluid communication with the supply tank and the distributor, the pump operable to pump the cleaning fluid in a direction from the supply tank to the distributor; and

a pump housing disposed within the housing, the pump disposed in the pump housing and the pump housing including a cooling air inlet and a cooling air outlet in fluid communication with the suction source, and

wherein the housing further includes a first tank guide projection that is received in the first recess of the supply tank when the supply tank is in the supply tank operating position, the first tank guide projection including a cooling air inlet in fluid communication with the cooling air inlet of the pump housing, and

wherein the housing further includes a second tank guide projection that is received in the second recess of the supply tank when the supply tank is in the supply tank operating position, the second tank guide projection forms a cooling air flow path between the cooling air inlet of the first tank guide and the cooling air inlet of the pump housing.

18. The surface cleaner of claim 17, wherein the cooling air inlet of the pump housing is located within the second tank guide.

19. The surface cleaner of claim 18, wherein the pump housing includes an inlet projection that includes the cooling air inlet of the pump housing, wherein the inlet projection extends into the second tank guide.

20. The surface cleaner according to any one of claims 17 through 19, wherein the cooling air flow path extends through the housing.

21. The surface cleaner according to any one of claims 17 through 20, wherein the housing includes a first side and a second side opposite the first side, wherein the first tank guide projection is disposed on the first side of the housing and the second tank guide projection is disposed on the second side of the housing.

22. A surface cleaner comprising:

a housing;

a suction nozzle coupled to the housing;

a recovery tank in fluid communication with the suction nozzle, the recovery tank disposed on the housing in a recovery tank operating position;

a suction source disposed in the housing, the suction source in fluid communication with the suction nozzle and the recovery tank, the suction source configured to draw fluid and debris through the suction nozzle and into the recovery tank when the recovery tank is in the recovery tank operating position;

a supply tank configured to store a cleaning fluid, the supply tank including a first recess and a second recess, the supply tank disposed on the housing in a supply tank operating position;

a fluid distributor in fluid communication with the supply tank when the supply tank is in the supply tank operating position, the distributor configured to dispense the cleaning fluid from the supply tank onto a surface being cleaned,

a pump in fluid communication with the supply tank and the distributor, the pump operable to pump the cleaning fluid in a direction from the supply tank to the distributor; and

a pump housing disposed within the housing, the pump disposed in the pump housing and the pump housing including a cooling air inlet and a cooling air outlet in fluid communication with the suction source, and

wherein the housing further includes a chamber having a first side and a second side opposite the first side, the chamber having a chamber floor at an elevation above the pump;

wherein the chamber further includes a chamber air inlet and a chamber air outlet in fluid communication with the cooling air inlet of the pump housing forming a cooling air path between the chamber air inlet and the pump housing.

23. The surface cleaner of claim 22, further comprising a cooling air flow passageway in communication with the chamber air outlet extending between the upper chamber and the pump housing providing a cooling air flow path between the upper chamber and the pump housing.

24. The surface cleaner of claim 23, wherein the cooling air flow passageway extends downwardly.

25. The surface cleaner of claim 23 or 24, wherein the cooling air flow passageway is aligned with the cooling air inlet of the pump housing.

26. The surface cleaner according to any one of claims 22 through 25, wherein the housing includes a first tank guide that facilitates locating at least one of the recovery tank in the recovery tank operating position and the supply tank in the supply tank operating position, wherein the first tank guide includes an air inlet in fluid communication with the chamber air inlet.

27. The surface cleaner according to any one of claims 22 through 26, wherein the housing includes a first tank guide that facilitates locating one of the recovery tank in the recovery tank operating position and the supply tank in the supply tank operating position, and a second tank guide that facilitates locating the other of the recovery tank in the recovery

tank operating position and the supply tank in the supply tank operating position, wherein the second tank guide forms the cooling air flow passageway.

28. The surface cleaner of claim 27, wherein the first tank guide projection is disposed adjacent the chamber first side and the second tank guide projection is disposed adjacent the chamber second side.

29. The surface cleaner according to any one of claims 22 through 28, wherein the cooling air flow path extends through the chamber.

30. The surface cleaner according to any one of claims 22 through 29, wherein the chamber air outlet is at an elevation above the chamber air inlet.

31. The surface cleaner according to any one of claims 22 through 30, wherein the chamber air outlet and the chamber floor are configured to inhibit liquid entering the chamber from traveling through the chamber air outlet.

32. The surface cleaner according to any one of claims 22 through 31, wherein the chamber floor is configured to direct liquid entering the chamber to exit through the chamber air inlet.

33. The surface cleaner according to any one of claims 22 through 32, wherein the chamber floor is positioned above the suction source.

34. The surface cleaner according to any one of claims 22 through 33, wherein the chamber air inlet is at an elevation above the suction source.

35. The surface cleaner according to any one of claims 23 through 34, wherein the cooling air inlet of the pump housing is located within the cooling air flow passageway.

36. The surface cleaner according to any one of claims 23 through 35, wherein the cooling air inlet of the pump housing is disposed on an inlet projection configured to inhibit liquid entering the housing from traveling through the cooling air inlet of the pump housing.

37. The surface cleaner of claim 36, wherein the inlet projection extends upwardly.